Status and evidence limits

Phases 1 and 2 are complete. Phase 3 is not started and is outside this snapshot.
Phase 1 is a 32-shot native-BP pilot_adaptation / not_assessed result, not an impulse-paper reproduction.
Phase 2 is a two-shot canonical_implemented / development_fixture / not_assessed result, not a source-faithful or statistical reproduction.
Profiling, tracing, and novelty entries remain provisional.
The verification reports, not this PDF, are the executable evidence authority.
This PDF is a portable rendering of the listed repository Markdown sources; source labels and SHA-256 digests below provide provenance.

Contents and source manifest

The following sections are rendered in this fixed order. Each digest is SHA-256 over the source Markdown bytes.

1. docs/phase1-source-decisions.md e07741064bc61d5861ff3ccc3826f546e9acb8bb72ebf27f0d73facf0e7ab8ed
2. docs/environment-pins.md f39637af63f04a8da4cf600fa3aa0455e3e3d04dc08213413198dcca1fb68f2f
3. docs/phase1-pilot.md bf16ac2a626b5344a17412c298be9aceaef1f7776c0afe65f120dda85e7bd8dc
4. .planning/phases/01-evidence-registry-and-native-css-baseline/01-VERIFICATION.md
4139c2c28f978ab81b2a451eec0e84bca493be4e94a58009e76e496903c89ad0
5. docs/snapshots/canonical-engine-through-phase-2.md
6173bad6a5996e92d33bdd407e854992fdd969e5332b7362da283d26fbf75a78
6. docs/profiling-and-tracing-strategy.md
02b0ca5ace775893e5b7bcb16bb3cec11d4eae09a143d24544cb753fd753f876
7. docs/research-novelty-and-depth-register.md
e6ce250cc1b28a9631e0ce4a45cbfc53e9a4d3051a2e58e95f54766aaa96064d
8. .planning/phases/02-canonical-css-execution-and-policy-contracts/02-VERIFICATION.md
13399018a4e323314f720921c5771a3d7b3970c6f86f8b4f746b7fc4552d2aeb

Phase 1 source decisions

Phase 1 source-artifact decision record

D-03 inventory

The inventory in `data/evidence/code-inventory.json` keeps three distinct leads. `B1 [[882,24,18<=d<=24]]` is a **lifted-product** code in the impulse paper (v1, Figure 5); it must not be conflated with the BPGD paper's C2. `C2 [[1922,50,16]]` is the **hypergraph-product** code named by the BPGD paper (v2, Figure 5). Gross/BB-144 is a bivariate-bicycle lead related at the parameter level to impulse Figure 6 and to the Bravyi construction paper.

Each lead records its paper relation separately from its executable source identity. A parameter tuple or family name never establishes matrix ordering, logical-basis row ordering, or source-faithful equivalence.

Conditional ranking before the source probe

1. **Gross/BB-144** was the conditional first choice because Bravyi's public repository supplied a full commit, `decoder_setup.py`, and a stated `bposd.css.css_code` logical-extraction path.
2. **B1 lifted product** had paper evidence but no verified original matrix or ordered-logical artifact.
3. **C2 HGP** had an explicit family relation but no pinned matrix seed, product ordering, or ordered-logical artifact.

The ranking was only a reason to probe Gross first. It was not a selection.

Gross pinned-source probe

The checked repository is `https://github.com/sbravyi/BivariateBicycleCodes` at `fa77e3333d3ec44c79d8f914dd24c040d1da471b` (2024-04-29). Its actual `decoder_setup.py` SHA-256 is `e919e1830b88468c5d257c0119656576a7999f92743dfeccd389a967524b3d2c`. The reviewed active parameters are `ell=12`, `m=6`, `a=(3,1,2)`, and `b=(3,1,2)`, with source ordering `Hx=[A|B]`, `Hz=[B^T|A^T]`.

`tools/freeze_css_source.py` reads that commit and reconstructs only this matrix recipe; it does not import the source program or run its circuit/data generation side effects. In the approved `.venv`, independent GF(2) calculation produced Hx and Hz shapes `(72,144)`, ranks `66` and `66`, `k=12`, and zero entries in `Hx Hz^T`. The approved `bposd==2.1` logical provider (`bposd/css.py` SHA-256 `d1e0c238d81c9b023bffa7bd355cfe305f8da3a6233a0af2edc1187f4807d6b1`) also reported `K=12` and emitted 12 Lx and 12 Lz rows.

The probe establishes Gross as the conditional first choice. Its provider rows are still not a selected artifact: Task 2 must record their actual row fingerprints, normalize the ordered Lz partners explicitly, and independently validate the resulting ordered paired basis.

Final D-04 decision and rejected candidates

**Gross/BB-144 is selected as the verified artifact for the first native

Phase 1 source decisions (continued)

pilot adaptation.** Task 2 regenerated the exact source rows and wrote `data/artifacts/selected-css.json` with raw SHA-256 `2c02a3b99bf7267acf35efe28e0a2ef0d3827e5a22b948d53da14ae8df68ceb0` and semantic SHA-256 `e175e67c0f9ae6f4e0968461ad2f87ba38ad5344166b793c53dd038976d4f427`. The provenance file preserves bposd's source Lx/Lz fingerprints and records the explicit GF(2) Lz pairing normalization; project-owned validation then checks commutation, ranks, quotient independence, and identity pairing.

B1 is rejected for this pilot because its exact matrix source and ordered logical rows are absent. C2 is rejected for this pilot because its source matrices, HGP construction ordering, and ordered logical rows are absent.

The selected code remains a **pilot adaptation**. Bravyi's repository is circuit-level BP-OSD software and the impulse Figure 6 lead is code-capacity; this shared code construction does not claim either source's decoding run or statistical curve has been reproduced.

Relation to the Phase 1 native pilot

No source-faithful reproduction or selected-code pilot is authorized by this record. The existing project-owned three-qubit run remains a disclosed pilot adaptation and does not fill this source-artifact gate.

Development environment

Observed Phase 1 development environment

Observed on 2026-09-22 UTC on `dcc-core-ferc-u-ac39-4-1.rc.duke.edu`.
These are local observations, not publication pins or a portability guarantee.
Repeat the probe on each execution node before interpreting a run.

Item	Observed value
Python	3.12.2, Anaconda build, GCC 11.2.0
Executable	`/hpc/home/mg551/miniconda3/bin/python`
Platform	Linux 5.14.0-687.29.1.el9_8.x86_64, glibc 2.34
NumPy	2.2.6, `/hpc/home/mg551/miniconda3/lib/python3.12/site-packages`
SciPy	1.17.1, same site-packages path
ldpc	2.4.1, same site-packages path
pytest	9.0.3, same site-packages path
pydantic	absent
bposd	absent

Distribution versions and paths were read with `importlib.metadata.distribution`.
No package was installed during this probe. `conda` emitted a
`conda-libmamba-solver` entry-point warning; the active Python worked.

Native API probe

With `OMP_NUM_THREADS=1`, `ldpc.BpDecoder` imported from
`/hpc/home/mg551/miniconda3/lib/python3.12/site-packages/ldpc/__init__.py`.
The constructor accepted `bp_method='product_sum'`, `schedule='parallel'`,
`input_vector_type='syndrome'`, `max_iter=10`, and `error_rate=0.1` for
`H = [[1,0],[0,1]]`. Decoding syndrome `[1,0]` returned candidate `[1,0]`,
`converge=True`, and `iter=1`. The C extension exposes no inspectable Python
constructor signature (`inspect.signature` raised `ValueError`).

The exact probe command was:

```bash
OMP_NUM_THREADS=1 python - <<'PY'
import numpy as np
from ldpc import BpDecoder
h = np.array([[1, 0], [0, 1]], dtype=np.uint8)
b = BpDecoder(h, bp_method='product_sum', schedule='parallel',
              input_vector_type='syndrome', max_iter=10, error_rate=0.1)
s = np.array([1, 0], dtype=np.uint8)
r = b.decode(s)
print(r.tolist(), bool(b.converge), int(b.iter))
PY
```

Approved isolated development environment

On 2026-09-22, after the Plan 02 human package gate, `.venv` was created with
`python -m venv --system-site-packages .venv`. The user approved exact PyPI
versions `pydantic==2.13.5` and `bposd==2.1`; those were installed with
`.venv/bin/python -m pip install 'pydantic==2.13.5' 'bposd==2.1'`.
Their resolved dependencies were `pydantic-core==2.46.5`,
`annotated-types==0.8.0`, `typing-extensions==4.16.0`, and
`typing-inspection==0.4.4`. Existing NumPy, SciPy, ldpc, and pytest were

Development environment (continued)

reused from the Anaconda site-packages directory and were not reinstalled.

`pydantic`, `pydantic-core`, and `bposd` import from
`/hpc/group/pfisterlab/mg551/storage/survey_bp/.venv/lib/python3.12/site-packages`;
NumPy, SciPy, ldpc, and pytest import from
`/hpc/home/mg551/miniconda3/lib/python3.12/site-packages`.
The full observed `pip freeze` is in `docs/environment-freeze.txt`.
Because this environment inherits system packages, reproduce it only on a
node with the same observed base package identities; a clean lock will be
needed before a portable release.

Phase 1 native CSS pilot

Phase 1 selected CSS pilot

The selected command runs the Plan 08 verified Gross/BB-144 artifact through one native `ldpc.BpDecoder` configuration: X-only code-capacity noise at `p=0.01`, product-sum updates, parallel schedule, maximum 30 iterations, and one OpenMP thread. It fixes 32 identified shots through a manifest-derived seed policy.

```bash
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli preflight \
  data/manifests/selected-css-pilot_adaptation-not_assessed.json
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli pilot \
  data/manifests/selected-css-pilot_adaptation-not_assessed.json \
  --output data/results/pilot_adaptation-not_assessed
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli verify-run \
  data/results/pilot_adaptation-not_assessed
```

The manifest binds the source revision, source-file SHA-256, frozen matrix and logical hashes, package versions, decoder settings, and finite budget before a native call. Flags cannot replace those scientific fields. The command rejects another BB artifact, source revision, environment, or decoder API contract.

Project-owned three-qubit tracer fixture

The original Phase 1 local tracer command is retained as a one-shot smoke fixture with a complete current manifest. It verifies the project-owned three-qubit CSS artifact, current environment, and native BP boundary before writing a begin and terminal event:

```bash
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.runner \
  data/manifests/project-three-qubit-fixture.json \
  --ledger /tmp/project-three-qubit-fixture-events.jsonl
```

The fixture is a `pilot_adaptation` with `not_assessed` output. Its linked paper gives only bit-flip methodological context; the code, logical basis, and one-shot budget are project-owned test data. It is neither a paper result nor a source-faithful reproduction.

Every displayed surface calls this a **pilot_adaptation** and **not_assessed**. The ledger and result preserve successful, nonconverged, invalid-syndrome, interrupted, and unavailable-outcome rows. `result.json` rebuilds denominators only from final ledger rows; rerunning fills only missing shot IDs and never double-counts final rows.

This is a local execution smoke pilot. It is not source-faithful reproduction, does not use the source impulse method, does not use OSD, and makes no curve pass or statistical-reproduction claim. The linked paper experiment remains `mapped_only` until later source-faithful qualification and assessment.

Phase 1 verification

---
phase: 01-evidence-registry-and-native-css-baseline
verified: 2026-09-22T04:33:43Z
status: passed
score: 31/31 must-haves verified
covered_files:
 - .planning/REQUIREMENTS.md
 - .planning/ROADMAP.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-01-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-01-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-02-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-02-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-03-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-03-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-04-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-04-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-05-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-05-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-06-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-06-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-07-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-07-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-08-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-08-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-09-PLAN.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-09-SUMMARY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-CONTEXT.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-DISCUSSION-LOG.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-PATTERNS.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-RESEARCH.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-REVIEW-FIX.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-REVIEW.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-SECURITY.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-SOURCE-AUDIT.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-VALIDATION.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/01-UAT.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/COVERAGE.md
 - .planning/phases/01-evidence-registry-and-native-css-baseline/SKELETON.md
 - data/artifacts/project-three-qubit-css-fixture.json
 - data/artifacts/selected-css.json
 - data/artifacts/selected-css.provenance.json
 - data/evidence/code-inventory.json
 - data/evidence/events.jsonl
 - data/evidence/protocol.json
 - data/evidence/seed-records.json
 - data/manifests/project-three-qubit-fixture.json
 - data/manifests/schema-v1.json
 - data/manifests/selected-css-pilot_adaptation-not_assessed.json
 - data/results/pilot_adaptation-not_assessed/events.jsonl
 - data/results/pilot_adaptation-not_assessed/result.json
 - docs/environment-freeze.txt
 - docs/environment-pins.md
 - docs/phase1-pilot.md
 - docs/phase1-source-decisions.md
 - pyproject.toml
 - src/bp_registry/__init__.py

Phase 1 verification (continued)

  - src/bp_registry/artifacts.py
  - src/bp_registry/cli.py
  - src/bp_registry/css.py
  - src/bp_registry/evaluator.py
  - src/bp_registry/evidence.py
  - src/bp_registry/ledger.py
  - src/bp_registry/manifests.py
  - src/bp_registry/reference.py
  - src/bp_registry/runner.py
  - src/bp_registry/sampling.py
  - tests/conftest.py
  - tests/test_css_small.py
  - tests/test_evidence.py
  - tests/test_ledger.py
  - tests/test_manifest.py
  - tests/test_phase1_cli.py
  - tests/test_reference_smoke.py
  - tests/test_sampling.py
  - tests/test_source_artifact.py
  - tools/freeze_css_source.py
covered_digest: "v1:sha256:c96be78e929c20e1c8e1cc71a8670b4ae55dcd664dc98e931833296185a6c412"
behavior_unverified: 0
overrides_applied: 0
unverified_prohibitions: 0
re_verification:
  previous_status: gaps_found
  previous_score: 27/31
  gaps_closed:
    - "Unsuccessful-row validation now rejects fabricated native, sample, and selection fields."
  gaps_remaining: []
  regressions: []
human_verification:
  - test: "Compare BPGD v2 and Bhatnagar impulse v1 CSS seed records with the pinned primary-source
  pages."
    expected: "Figure/table anchors, settings, extracted claims, and documented unknowns agree with the
    sources."
    why_human: "Editorial source interpretation cannot be established by internal code and fixtures."
  - test: "Compare frozen Gross matrix/logical rows with the pinned Bravyi checkout and bposd 2.1
  provider bytes."
    expected: "Upstream rows, construction order, logical-provider output, and recorded digests agree."
    why_human: "The upstream checkout/source bytes are not shipped with the phase artifact."
  - test: "Confirm the Plan 02 package legitimacy decision against the original human checkpoint
  record."
    expected: "The exact pydantic and bposd package/version approval preceded installation."
    why_human: "The codebase retains a summary of approval, not independent human provenance."
  - test: "Review the three flagged scientific-claim prohibitions in Plans 06 and 09."
    expected: "The pilot is never presented as source-faithful or statistically reproduced, and
    unresolved identities are not promoted to facts."
    why_human: "These judgment-tier prohibitions were explicitly left unverified for end-of-phase
    review."

Phase 1: Evidence Registry and Native CSS Baseline Verification

Phase goal: As a researcher, I want to trace a source-linked paper record to a validated native CSS experiment, so that I can inspect its outcomes and provenance.

Phase 1 verification (continued)

Status: `passed` — 31 of 31 PLAN truths verified, none failed. The four human checks passed in `01-UAT.md`; the three scientific-claim prohibitions were included in that review. This follows re-verification after commits `0cdf1a7`, `d0b88e9`, and `2022257`. The fixture, unsuccessful-shot counting, and T-01-21 provenance gaps are closed. The five ROADMAP success criteria were checked independently and map to the PLAN truths below.

User Flow Coverage

Step	Expected	Code and observed evidence	Status
Find paper record	Open a revision-linked result and identify its evidence maturity	`seed-records.json` has Bhatnagar v1 Figure 6 and `mapped_only` paper experiment; `evidence.load_seed_records` and `validate-evidence` pass	VERIFIED
Resolve experiment	Bind selected code, logicals, native settings and source identities	Manifest, provenance and preflight bind Gross/BB-144 artifact to Bravyi commit and `ldpc==2.4.1`; public preflight passed	VERIFIED for adaptation; source-paper fidelity needs human review
Run reference	Invoke documented local native BP command for 32 identified shots	`runner.run_one` calls `reference.decode(Hz, sZ, prior...)`; checked-in journal has 32 begin and 32 terminal events	VERIFIED for checked-in run
Inspect outcomes	Reconstruct counts and provenance for the executed experiment	`verify-run` passed on all 32 completed rows; named tests count interruption/exception and reject fabricated unavailable fields	VERIFIED in code; source interpretation needs human review

The Gross construction is a verified local code artifact for this **pilot adaptation**. It is only a parameter-level lead for the Bhatnagar impulse Figure 6 experiment. The checked-in run is BP-only, uses 32 fixed shots, and is `not_assessed`. The Bhatnagar paper experiment stays `mapped_only`; the distinct Gross pilot event is `reference_executable`. This is not a source-faithful impulse run or a statistical curve reproduction.

Roadmap Success Criteria

#	Contract	Status	Evidence
1	Versioned mapping protocol and honest seed corpus	VERIFIED	`protocol.json`, `seed-records.json`, event replay and taxonomy queries; source statements still await editorial comparison.
2	Complete versioned manifest and immutable preflight	VERIFIED	Strict `ManifestV1`, `artifacts.load_artifact`, environment/hash checks, and invalid-input tests.
3	CSS algebra, ordered identity, sampling and four outcomes	VERIFIED	`CssCode.validate`, semantic bytes, `sample_shot`, independent evaluator and constructed-case tests.
4	Documented pinned native BP pilot with inspectable outcomes	VERIFIED	`verify-run` returned 32 final rows and `not_assessed`; named tests cover interruption/exception denominators and tampering rejection.
5	Distinct evidence maturity states	VERIFIED	Paper is `mapped_only`; linked pilot is `reference_executable`, both `not_assessed`; validator rejects paper promotion.

PLAN Must-Haves: Observable Truths

ID	Truth condensed from PLAN frontmatter	Status	Evidence or limit
01-01.1	Wave 0 test collection precedes later tests	VERIFIED	Git history places `183ee7c`/`f745d27` Wave 0 tests before later phase implementation commits.
01-01.2	Actual node package identities recorded and rechecked	VERIFIED	`environment-pins.md`; manifest preflight compares live Python, NumPy, SciPy, ldpc and OpenMP values.
01-02.1	Human examined suspicious package identities before installation	UNCERTAIN	`01-02-SUMMARY.md` and `environment-pins.md` record a user confirmation, but neither independently

Phase 1 verification (continued)

proves the human review. Preserve the checkpoint record for human sign-off. |
01-03.1	Three-qubit fixture runs full local tracer path	VERIFIED	Complete fixture manifest and frozen project artifact preflight; documented `bp_registry.runner` command exited 0 and wrote begin/terminal rows.
01-03.2	Fixture is labelled smoke/adaptation, never a paper result	VERIFIED	Fixture manifest and docs disclose project-owned code, `pilot_adaptation`, and `not_assessed`.
01-03.3	Begin precedes sampling and terminal links native/evaluation fields	VERIFIED	`runner.run_one` ordering and 64 checked-in events; `test_runner_consumes_the_verified_resolved_experiment` passed.
01-04.1	Validate CSS dimensions, GF(2) commutation, rank, logical independence/pairing	VERIFIED	`css.py:CssCode.validate`; small-code and selected-artifact tests passed.
01-04.2	Role/order/bit-sensitive raw and semantic hashes	VERIFIED	`css.semantic_bytes`, `manifests.matrix_hashes`, mutation/golden-byte tests passed.
01-04.3	X-only and separate X/Z samples with boundaries/rate rejection	VERIFIED	`sampling.sample_shot` and `tests/test_sampling.py` passed.
01-04.4	Four exclusive outcomes from original constraints and hidden truth	VERIFIED	`evaluator.evaluate_candidate`; constructed exact, degenerate, logical and invalid tests passed.
01-04.5	Version 1 artifact loader remains explicit	VERIFIED	`artifacts.load_artifact` dispatches v1 and rejects unknown versions without rewriting bytes.
01-05.1	One durable terminal per begun shot, including failure/interruption	VERIFIED at ledger layer	`LedgerWriter` and `test_active_owner_stays_pending_then_dead_owner_is_repaired_once` passed. Public pilot consumption is a separate failed truth.
01-05.2	Empty/adjacent/duplicate journal behavior	VERIFIED	`reconcile`, `_states`, and named ledger tests passed.
01-05.3	Concurrent shot ownership and final-row-only counts	VERIFIED	Owner locks, `summarize`, and active-owner test passed.
01-05.4	Reconcile only abandoned owners	VERIFIED	Live owner produces no final row; released owner produces one interrupted row in named test.
01-06.1	BPGD/impulse CSS extraction matches source revisions	UNCERTAIN	Records enumerate anchored figures and tables with unavailable curve data disclosed; independent comparison to pinned paper pages is the declared manual source-fidelity check.
01-06.2	B1, C2 and Gross remain distinct code leads	VERIFIED	`code-inventory.json` records lifted-product B1, HGP C2 and BB Gross with different identity maturity.
01-06.3	Search/screen/revision history retains order and empty/adjacent cases	VERIFIED	`EvidenceRegistry` and `test_search_import_preserves_adjacent_raw_hits_and_empty_search` passed.
01-06.4	Repeated/concurrent events and claim maturity are handled	VERIFIED	Content-addressed append, lock, conflict and taxonomy tests passed.
01-07.1	Complete versioned manifest fields	VERIFIED	Selected manifest and strict Pydantic models include decoder, hashes, noise, seed, budget, environment, expected-output state and discrepancies.
01-07.2	Invalid/missing/escaped/changed inputs fail before native call	VERIFIED	`preflight` orders validation before runner sampling; spy and path/hash mutation tests passed.
01-07.3	Unassessed pilot has reason	VERIFIED	Manifest, CLI and result carry `not_assessed` and reason.
01-07.4	Saved v1 records remain parseable via version dispatch	VERIFIED	Complete v1 parses; legacy v1 can be inspected through `_parse_for_preflight`, though it cannot execute.
01-08.1	Inventory distinguishes three candidate families and unresolved identities	VERIFIED	Candidate records and `test_inventory_distinguishes_b1_c2_and_gross_and_records_d04_gate` passed.
01-08.2	Selection uses source/artifact/logical/algebra evidence	VERIFIED	Freeze recipe checks pinned commit/source construction; artifact rows and GF(2) checks are tested independently.
01-08.3	Exact upstream Gross source and logical rows are independently confirmed	UNCERTAIN	Provenance has commit, source-file and provider digests, but no pinned checkout/source bytes are shipped to rerun the source probe here; manual source comparison remains required.
01-08.4	Frozen raw and semantic hashes reflect actual files	VERIFIED	`load_artifact` recomputes both hashes; checked-in artifact and provenance agree.
01-09.1	Selected BP-only pilot command and honest labels	VERIFIED	`docs/phase1-pilot.md`, `preflight` and `verify-run` commands passed; manifest/result label adaptation and no assessment.

Phase 1 verification (continued)

01-09.2	One manifest resolves 32 linked samples and result	VERIFIED	32 distinct shot IDs, 64 events, complete result, pinned versions and hashes; `verify-run` passed.
01-09.3	Every selected outcome, including unsuccessful/interrupted, reaches trustworthy denominators	VERIFIED	Named tests prove interrupted/exception counts; hardened `_validate_pilot_ledger` rejects fabricated candidate, sample, native work and selection fields (T-01-21 closed).
01-09.4	Pilot maturity does not promote paper experiment	VERIFIED	`load_seed_records` keeps Bhatnagar Figure 6 `mapped_only`, pilot `reference_executable`; promotion test passed.

Score: 31/31 verified; 0 failed; 0 uncertain after the four human checks in `01-UAT.md`. The prior fixture, counting, and T-01-21 gaps are now closed. No accepted overrides exist.

Advisory (New Scope, Unevidenced)

None. The previously blocking T-01-21 finding has a passing tampered-row rejection test and direct rejection spot-check.

Required Artifacts and Key Links

Artifacts or link	Existence and substance	Wiring and data flow	Status
Protocol, seed records, event journal, inventory	Present and populated with source IDs/anchors, status, candidate gaps	`EvidenceRegistry` loads events and claims; `validate-evidence` resolves pilot result	VERIFIED; source accuracy needs human review
CSS artifact and source provenance	Present with 72×144 checks and 12 ordered logical pairs	Manifest hashes feed `artifacts.load_artifact` then `CssCode.validate`	VERIFIED locally; external source fidelity uncertain
Manifest schema and preflight	Complete v1 model plus exported schema	CLI/pilot call preflight before sampling/native decoder	VERIFIED
Native adapter and evaluator	Real `ldpc.BpDecoder` call; four-way independent evaluator	`runner.run_one` passes only `Hz`, syndrome and prior to decoder; hidden truth reaches evaluator	VERIFIED
Durable ledger and result	32 begin/32 completed terminal events; result has 32 shot IDs	CLI reconstructs counts from final rows; named tests count interruption/exception and reject fabricated unsuccessful-row fields	VERIFIED
Three-qubit fixture manifest → one-shot runner	Complete manifest and project-owned artifact exist	Documented command preflights, decodes and writes begin/terminal events	VERIFIED
Registry paper record → pilot result	Distinct paper/pilot IDs and statuses	`load_seed_records` invokes `verify_run` and checks reciprocal run ID	VERIFIED

No rendered UI values exist in this CLI phase. The data-flow trace is: pinned seed record → manifest and selected artifact → preflight → deterministic sampler → native candidate → independent evaluator → terminal JSONL → result reconstruction → registry pilot event. Interrupted and exception rows reach the result with explicit unavailable fields that the validator now checks.

Behavioral Spot-Checks and Test Quality

Check	Observed result	Status
`OMP_NUM_THREADS=1 .venv/bin/python -m pytest -q`	71 passed in 25.13 s	PASS
Documented `bp_registry.runner` fixture command	Exit 0; one begin and one terminal, `outcome=exact`, `pilot_adaptation`, `not_assessed`	PASS; prior gap closed
Documented pilot `preflight`, `pilot`, and `verify-run` commands	Each exited 0; 32 final rows, `pilot_adaptation`, `not_assessed`	PASS
`test_public_pilot_counts_reconciled_interruption`	1 passed; 32 denominator, one interrupted and unavailable outcome	PASS; prior count gap closed
`test_public_pilot_counts_recorded_native_exception`	1 passed; 32 denominator, one exception and

Phase 1 verification (continued)

unavailable outcome | PASS; prior count gap closed |
| In-memory interrupted terminal with fabricated `candidate`, `native_iterations`, `native_edge_updates`, `selection_status` passed to `_validate_pilot_ledger` | Rejected: `pilot unsuccessful terminal lacks explicit unavailable fields` | PASS; T-01-21 closed |

There are no declared `probe-*.sh` scripts. Requirement-linked tests contain no disabled or skipped cases. Small-code algebra and native syndrome tests use value assertions; fixture-source regeneration checks rows and hashes, but paper/source agreement remains a manual editorial check. The interruption test now mutates unavailable native/sample fields and asserts public `verify-run` rejects them. No unreferenced `TBD`, `FIXME`, or `XXX` debt markers were found in phase implementation files. `01-REVIEW.md` listed eight findings; their fixes remain present. The updated `01-SECURITY.md` records T-01-21 closure and 24/24 threats closed; the direct code/test checks above support that finding.

Requirements Coverage

IDs	Status	Evidence and limit
MAP-01, MAP-02, MAP-03	SATISFIED in code; source-fidelity human review pending	Protocol, event history, pinned seed claims and queryable taxonomy.
REG-01, REG-02	SATISFIED	Strict manifest and preflight reject invalid schema, paths and identities.
REG-03	SATISFIED	Native `ldpc` reference executes from both the complete selected manifest and the restored fixture manifest.
REG-04	SATISFIED	Paper `mapped_only`, pilot `reference_executable`, both `not_assessed`.
CSS-01, CSS-02, CSS-03, CSS-04	SATISFIED locally	GF(2) and hash tests, identified sampling and four-outcome evaluator.
LED-01	SATISFIED in code	Public result counts failures and interruptions and rejects fabricated unavailable native fields on those terminal rows.

All 12 ROADMAP Phase 1 requirement IDs appear in PLAN frontmatter; none is orphaned. Statistical curve assessment and broader campaigns are explicitly later-phase work. The current 32-shot pilot remains `not_assessed`.

Decision Coverage

`gsd-tools check decision-coverage-verify` reported 5 of 5 CONTEXT decisions honored, with no unhonored entries. This is advisory and does not resolve the source-fidelity human checks.

Human Verification Required After Gap Closure

1. Compare BPGD v2 and Bhatnagar impulse v1 seed claims, figure/table anchors, settings and unresolved fields to the pinned primary sources. Record any editorial correction. This resolves 01-06.1 and the mapping-source limitation.
2. Compare the frozen Gross rows and ordered logical-provider output with the pinned Bravyi commit and `bposd==2.1` implementation bytes. The local tests confirm internal algebra, not upstream source identity. This resolves 01-08.3.
3. Confirm the Plan 02 exact-version package legitimacy decision from the original human checkpoint record. The summary reports approval; independent human provenance is not in the codebase.
4. Review the three flagged judgment-tier prohibitions in Plans 06 and 09: do not promote a pilot to paper reproduction, do not call unresolved identities proven absent/source-faithful, and do not infer source fidelity from matching code parameters. Current labels are conservative, but the plan flags require explicit human resolution.

Gaps Summary

The selected Gross/BB-144 BP-only pilot is a traceable 32-shot **pilot adaptation**, with all checked-in

Phase 1 verification (continued)

 shots completed and its published source experiment still `mapped_only`. All automated gaps are closed.
 Source-fidelity, the historical package checkpoint, and the three flagged scientific-claim judgments remain for the end-of-phase human checkpoint. No statistical curve reproduction is claimed.

Verified: 2026-09-22T04:33:43Z
Verifier: gsd-verifier

Canonical CSS engine

Canonical CSS engine: a checked development tutorial and contract reference

This guide runs the project-owned binary CSS sum-product implementation. It is not an adapter to a native decoder, a source-faithful reproduction, or a statistical result. The checked fixture is explicitly labelled `canonical_implemented`, `development_fixture`, and `not_assessed`.

Scope and approved environment

Use the approved `.venv` described in environment-pins.md. It is a node-specific development environment, not a portable release lock. The commands below assume the repository root and a checked working tree. They do not fetch packages, alter the checked result, or expose hidden error truth to the decoder.

If an approved environment needs its local project import registered, the only permitted editable-install command is:

```bash
.venv/bin/python -m pip install --no-deps -e .
```

`--no-deps` is intentional: it prohibits dependency fetching. Do not run `pip install`, upgrade a package, or recreate `.venv` as part of reproducing this tutorial. During Phase 2 validation, the approved environment already imports the checked project source without running that command.

Run, inspect, reconcile, and verify the checked experiment

The public manifest is `data/manifests/canonical-three-qubit-development.json`. First, preflight it:

```bash
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli preflight
data/manifests/canonical-three-qubit-development.json
```

Preflight validates the schema, environment declaration, artifact and matrix identities, closed policy bindings, profile, and budgets before a sample, ledger, or kernel operation. Its JSON includes `run_id`, `numerical_profile`, the resolved `policy_identity`, `shot_budget`, and the development-evidence labels.

Run an isolated, disposable copy so the checked result is never modified:

```bash
OUT="$(mktemp -d "${TMPDIR:-/tmp}/canonical-three-qubit-development.XXXXXX")"
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli canonical
data/manifests/canonical-three-qubit-development.json --output "$OUT"
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli reconcile "$OUT/events.jsonl"
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli verify-run "$OUT"
```

The `canonical` report and `verify-run` report both return `final_row_count: 2`. The reconciliation query reports two final rows. The terminal results for this deterministic fixture are `completed`; their

Canonical CSS engine (continued)

independently evaluated outcomes are `logical_failure` for shot 0 and `exact` for shot 1. `verify-run` recomputes the result from the hash-linked structural ledger, checks the manifest identity and independent evaluation, and rejects mismatched or relabelled evidence.

Inspect the persisted, checked result without executing a shot:

```bash
OMP_NUM_THREADS=1 .venv/bin/python -m json.tool
data/results/canonical-three-qubit-development/result.json
OMP_NUM_THREADS=1 .venv/bin/python -m bp_registry.cli verify-run
data/results/canonical-three-qubit-development
```

Expected result fields map to contracts as follows:

Result or event field	Contract it identifies
`run_id`, `manifest_sha256`, `ledger_sha256`	declared experiment and hash-linked durable evidence
`policy_identity`	resolved, ordered seven-role composition
`relation`, `fixture`, `assessment_status`, `statistical_claim`	evidence maturity: canonical implementation, development fixture, not assessed
terminal `outcome`, `logical_action_agrees`, `termination`	evaluation after selection and the terminal reason
`qualified_iterations`, `work_vector`	schedule-qualified flooding work, not a universal cost scalar
terminal `profile` and `runtime` profile evidence	typed per-attempt CPU runtime/resource facts, adapter overhead, and explicit unavailable backend-device facts
`original_lineage_id`, `working_lineage_id`, `candidate_lineage_id`	immutable ancestry from problem through frozen candidate
`trace_plan`	public tracing is admitted only as `off` until a canonical trace hook can emit bounded sidecars honestly

The checked fixture has trace level `off`; consequently it records no trace sidecar digest. That absence is declared by the manifest rather than inferred from an empty trace file.

Mathematical and numerical conventions

The persisted public manifest profile is `cpu-float64-spa-v1`; preflight materializes its resolved engine profile as `binary-spa-flooding-f64-v1` in `resolved_numerical_profile`. That engine profile is materialized by `src/bp_registry/canonical/numerics.py` and `src/bp_registry/canonical/schedules.py`. It uses deterministic CPU NumPy `float64` and natural-log likelihood ratios,

$$
L = \log(P(0) / P(1)).
$$

For X-error recovery in the binary CSS component, the observed Z-check syndrome convention is $s_Z = H_Z e_X \pmod 2$, with bit vectors ordered by the immutable CSS artifact. The canonical sampler therefore forms the decoder-visible syndrome from H_Z and the hidden X-error vector e_X; it does not substitute H_X for this recovery component. A hard-decision tie is resolved deterministically by the profile's canonical

Canonical CSS engine (continued)

index/order rule; no physical-error truth or unrecorded random draw chooses a tie.

The check update uses the stable LLR-domain SPA (box-plus) procedure: combine incoming messages through the sign/parity and stable magnitude operation, then apply the observed syndrome sign. Variable-to-check values are the channel LLR plus the other incoming check messages. The profile clips finite calculation boundaries symmetrically at `K = 25`. A non-finite message is a recorded `NumericalFailure` terminal condition, not a silently propagated value. Degree-one checks, zero messages, normalization, initialization, and ties have deterministic tested handling in the numerical profile; an alternative precision, clipping limit, or randomized tie rule must be a new named profile or declared policy, never an implicit substitution.

Graph order and flooding buffers

Sparse input enumeration is normalized to a canonical **check-major, variable-minor** edge table. One flooding sweep reads the complete **old v2c snapshot** (the old v2c snapshot), computes all check-to-variable messages, forms beliefs, and only then commits the complete **fresh v2c buffer** (the fresh v2c buffer). It never interleaves fresh
and stale variable-to-check values. The atomic work unit is therefore a complete double-buffered sweep, not an individual array assignment.

The source anchors are `CanonicalEdgeTable` and `FloodingSchedule.sweep` in `src/bp_registry/canonical/schedules.py`, `BinarySpaKernel` in `src/bp_registry/canonical/kernel.py`, and the finite boundaries in `src/bp_registry/canonical/numerics.py`. These are project procedure anchors; they do not claim that a paper's implementation used identical numerical choices.

Seven explicit policy roles and lifecycle

Every persisted `CanonicalExperimentV2` names every role, including explicit identity policies when inactive:

Role	Checked fixture binding	Responsibility
`update`	`binary_spa`	finite binary SPA message procedure
`schedule`	`flooding`	edge/message ordering within an attempt
`state_memory`	`stateless`	declared policy-local memory behavior
`intervention`	`identity`	declared no-op search/intervention behavior
`stopping`	`syndrome_or_budget`	syndrome/budget termination decision
`selection`	`stable_first`	deterministic candidate choice
`postprocessing`	`identity`	declared no-op postprocessing behavior

The engine, rather than a policy callback, traverses the named hooks at **shot, search, member, leg, branch, attempt, iteration, and message** scope. A policy receives a capability-scoped decoder-visible view, owns only declared private state, and returns typed decisions or mutations that the engine validates. It cannot obtain hidden sampled truth, evaluator state, ledger ownership, or arbitrary kernel internals. Binding evidence records a procedure basis and ordered invariants alongside the closed registry entry.

Compatibility is enforced twice. **Static composition validation** rejects unknown kinds, omitted roles, unsupported declarations, and incompatible

Canonical CSS engine (continued)

binding metadata before sampling. A **live-state invariant** check protects runtime boundaries. A source semantic that cannot be expressed through these declared hooks needs a versioned lifecycle extension or a documented composite controller, rather than an undocumented global callback order.

Budgets, work, runtime, lineage, and evaluation

Count budgets are hierarchical hard caps at attempt, member/leg, search, and shot scopes. The engine must **reserve** the full proposed work before a state-changing sweep and commits it only after the fresh buffer is finite. Wall-clock limits are separate boundary checks, not retroactive work counts. `WorkVector` records leaf work once (including c2v/v2c updates, belief assemblies, candidate checks, and other typed units); ancestor totals are derived through lineage instead of double-counted.

`iterations` is intentionally retained as a familiar headline but is a **schedule-qualified iteration**: here, a flooding complete-double-buffered sweep. Per-attempt runtime/resource evidence is emitted as a typed local profile: wall and process CPU time, peak host memory, adapter overhead, and explicitly unavailable backend-device timing/memory for this CPU fixture. Disabled profiling emits typed `not_collected` missingness rather than zeroes; the profile values are observational and never affect decoding, stopping, or candidate selection.

Lineage IDs are derived from typed logical parents and canonical content, not event arrival, wall time, or process identity. Candidate retention is a storage choice after identity, while selection uses declared stable order. The selected payload becomes a **frozen decoder-visible candidate** before the independent evaluator receives the hidden sampled truth. The evaluator then judges the original CSS problem and appends an evaluation annotation; it cannot change selection. This separation is why an exact development outcome is not evidence of source-faithful or statistical reproduction.

Compatibility examples

- The checked composition is compatible because it names all seven closed bindings and uses the CPU `float64` flooding profile.
- Replacing `flooding` with a residual scheduler is not a harmless label change: it needs a registered schedule procedure, its own qualified iteration semantics, and compatible lifecycle/work evidence.
- A random tie rule must declare replayable decoder-local randomness independent of physical-error sampling; it cannot reuse the deterministic profile name.
- A relay leg may transfer only a typed, named payload and has sequential attribution. It cannot inherit arbitrary predecessor private state.

Evidence boundary

This two-shot result is a bounded development smoke fixture. It supplies an executable contract check, not a publication curve, a population estimate, or a claim of established methodological novelty. Keep source-faithful reproduction and controlled comparisons as separately declared experiment relations with their own manifests, prespecified denominators, tolerances, and evidence.

Profiling and tracing

Profiling and tracing strategy (D-20)

Purpose, scientific value, and cost

Profiling and tracing make a canonical execution auditable when a result needs more explanation than an outcome and a headline iteration count. They are observations, not decoder controls: they must not change a candidate, work vector, lineage, termination, or selection. Their scientific value is to separate algorithmic work from process-local runtime, make replay agreement testable, and preserve bounded diagnostic evidence. Their cost is runtime, memory, I/O, and potential observer perturbation; measure that cost explicitly instead of folding it into clean execution time.

This is a development-only strategy, not a campaign default. It gives no publication quota, sampling rate, statistical tolerance, or compute allocation. Publication campaigns must declare their own values rather than inheriting Phase 2 defaults.

Evidence levels and bounded traces

All canonical runs retain sparse structural evidence: durable begin, attempt, candidate, selection, work, runtime, missingness, and termination facts. This is the minimum evidence needed to reconstruct a result. Optional traces add iteration and edge observations as sidecars.

Level	Implemented control	Use and limitation
off	`TracePlan.level="off"`	no sidecar; structural evidence remains
full	`TracePlan.level="full"`	every requested optional observation, only for small diagnostic development cases
filtered	`TracePlan.level="filtered"`, `iterations`, `edges`	selected iteration or edge evidence, for rare or race-sensitive investigation

`TracePlan` confines `relative_dir` under its declared root. `max_records` and `max_bytes` are hard quotas. A sidecar carries `truncated` and `dropped_records`; the truncation fact itself is retained even if prior optional records must be removed to fit the byte cap. Sidecars are content-addressed by SHA-256 and linked with path, digest, record count, byte count, and truncation status. The semantic trace digest is content identity for replay comparison: it removes measurement-only fields and is not authenticity or a population measurement.

Low-overhead bounded traces are particularly appropriate for race-sensitive behavior that cannot safely be reproduced by a second run. They remain observational: a trace sink can drop optional records but has no decoder control-flow interface.

Local profiling facts and missingness

`ProfilingPlan` defaults to enabled `low_overhead` observation; it can be disabled, made strict with `required`, or raised to `representative` with `track_python_allocations`. `LocalProfiler` emits typed `ProfileMeasurement` evidence: `runtime.wall_time` from `time.perf_counter_ns`, `process_cpu_time`, platform-qualified `peak_host_memory`, concurrency, machine adapter identity, and explicitly unavailable backend device facts; `adapter_overhead`, measured with an empty adapter call and reported apart from observed run duration; and `python_allocation_peak` from `tracemalloc` only when requested, not host RSS.

Only positive numeric thread counts from `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, `MKL_NUM_THREADS`, and `NUMEXPR_NUM_THREADS` are retained. Every unavailable measurement has method, scope, unit, precision, and a reason. Do not replace unavailable data with zero, guess child-process or device measurements, or serialize arbitrary environment values. In a strict plan, an unavailable required fact raises `ProfilingRequirementError` rather than quietly satisfying the request.

Profiling and tracing (continued)

Cohorts and what can be estimated

Use three separate cohorts, never a single selected example for every claim:

1. **clean**: the full clean population supplies outcome and ordinary runtime summaries, with `selection_reason="full_clean_population"`.
2. **representative**: a prespecified, paired clean/profiled sample for detailed profiling and overhead estimation. It records `selected_before_outcome`, stratum, and `inclusion_probability` before outcomes; only this form can be `population_estimate_eligible`.
3. **diagnostic**: targeted tail, rare-event, divergence, or bounded-trace replays for explanation. It records a selection reason and is not eligible for population inference.

Diagnostic evidence is not a population estimate. Outcome-conditioned or post-hoc selection must not be plotted as an unbiased population profile unless its selection probabilities and a valid weighting plan are declared separately.

Pairing, replay, agreement, and divergence

Create a `ReplayLink` between clean and profiled attempts with distinct `clean_run_id` and `profile_run_id`, plus `manifest_identity`, `sample_identity`, `decoder_randomness_identity`, `ordering_identity`, cohort, profiling level, and `overhead_ns`. Compare `semantic_agreement` only over `candidate`, `termination`, `work`, `lineage`, and `semantic_trace_digest`; timing and other measurement fields are deliberately excluded from semantic identity.

An agreement is necessary, but not by itself sufficient, for a representative population statement. A mismatch is retained as `diagnostic_divergence` with `retained_divergence` retention. It must never be silently dropped or described as a measurement of the clean execution.

Plotting and reporting

For every plot, state its cohort, denominator, profile level, sampled strata, inclusion probability when applicable, timing scope, and missingness counts. Use separate panels or marks for clean population outcomes/runtime, paired representative profile measurements and adapter overhead, and targeted diagnostic examples. Show truncated trace counts and unavailable counters; do not impute a zero or draw an unlabelled aggregate.

Development evidence is not a reproduction claim. A plotted two-shot fixture, selected trace, or profiler example is labelled `development_fixture` and `not_assessed` unless a separate source-faithful/statistical protocol establishes otherwise.

Retention and review

Retain `paired_summary`, `bounded_trace`, or `retained_divergence` explicitly. Keep selected candidates in full; rejected payloads require a bounded trace, rare-event cohort, or declared candidate-retention policy. Preserve hashes, lineage, reasons, validations, sidecar links, and missingness facts even when a large optional payload is dropped. Revisit quotas, cohort rules, and plotting templates before any campaign with allocated compute resources.

Implementation anchors: `src/bp_registry/canonical/profiling.py` (`ProfilingPlan`, `ProfileMeasurement`, `CohortSelection`, `ReplayLink`) and `src/bp_registry/canonical/tracing.py` (`TracePlan`, `TraceSink`, `semantic_digest`).

Research novelty and depth

Research novelty and depth register (D-26)

This is a living register, not a novelty claim. It connects each potential contribution to executable evidence and records what still needs a closest precedent and corpus recheck. Claim maturity is intentionally conservative: every entry is not established methodological novelty, and the development fixture remains `not_assessed`.

Potential contribution	Executable evidence	Closest precedent / current basis	Unresolved search	Claim maturity
Canonical policy composition for binary CSS BP	`src/bp_registry/canonical/contracts.py`, `execution.py`, seven-role manifests, `tests/canonical/test_contracts.py`	Existing BP implementations commonly fix update/schedule choices; this project’s evidence is a procedure contract, not a claim of first abstraction	Map decoder-policy frameworks and QLDPC BP variants for comparable explicit role/lifecycle schemes	implementation-backed design; provisional
Typed work, budgets, lineage, and independent evaluation boundary	`budgets.py`, `lineage.py`, `execution.py`, work/lineage tests and canonical structural ledger	Work accounting and provenance are established concerns; exact closest QLDPC decoder evidence register remains incomplete	Search QLDPC reproduction/provenance and work-normalization practice	implementation-backed design; provisional
Local profiling with representative pairing	`profiling.py`, `tests/canonical/test_profiling.py`, D-20 strategy	Google-Wide Profiling and stratified-sampling profiling motivate overhead and sampling separation	Recheck decoder-benchmark methodology and whether an equivalent clean/profile semantic-pair contract exists	potential contribution; provisional pending corpus recheck
Selective tracing with content-addressed sidecars	`tracing.py`, `tests/canonical/test_tracing.py`, D-20 strategy	Event tracing and content hashes are established engineering patterns; this project has no verified closest decoder-specific selective-trace precedent yet	Search decoder debugging/tracing, replay, and rare-event evidence practices	potential contribution; provisional pending corpus recheck

How to use the register

- **Potential contribution** states the narrow method or artifact, not a superlative. It must be traceable to executable evidence and tests.
- **Closest precedent** records a cited or explicitly incomplete comparison. “No verified closest precedent yet” means unresolved search, not novelty.
- **Unresolved search** is carried forward into the mapping study and must be refreshed before an abstract, manuscript, or review asserts differentiation.
- **Claim maturity** may advance only after corpus recheck, precise comparison, and evidence appropriate to the proposed claim. Development output is not assessed reproduction, population performance, or established methodological novelty.

Locked evidence limitations

Profiling and selective tracing are deliberately listed as provisional. A representative paired cohort can support a specified profiling estimate only when selection precedes outcomes, inclusion probability is retained, and semantic agreement is established. Targeted diagnostic replays and bounded traces remain explanatory evidence, not population evidence. The canonical three-qubit development result is canonical implementation evidence, not a source-faithful reproduction or a statistical reproduction claim.

Before manuscript use, re-run the unresolved search against the expanded literature corpus, record the exact compared implementations and definitions, and downgrade or remove any statement that lacks a defensible closest precedent.

Phase 2 verification

---
phase: 02-canonical-css-execution-and-policy-contracts
verified: 2026-09-24T01:24:49Z
status: passed
score: 5/5 must-haves verified
covered_files:
- .planning/PROJECT.md
- .planning/REQUIREMENTS.md
- .planning/ROADMAP.md
- .planning/STATE.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-01-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-01-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-02-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-02-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-03-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-03-SUMMARY.md
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- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-05-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-06-PLAN.md
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- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-07-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-07-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-08-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-08-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-09-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-09-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-10-PLAN.md
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- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-11-PLAN.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-11-SUMMARY.md
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- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-15-SUMMARY.md
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- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-16-SUMMARY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-CONTEXT.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-REVIEW.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-SECURITY.md
- .planning/phases/02-canonical-css-execution-and-policy-contracts/02-UAT.md
- data/artifacts/project-three-qubit-css-fixture.json
- data/manifests/canonical-three-qubit-development.json
- data/manifests/schema-canonical-v2.json
- data/results/canonical-three-qubit-development/events.jsonl
- data/results/canonical-three-qubit-development/result.json
- docs/canonical-engine.md
- docs/profiling-and-tracing-strategy.md
- docs/research-novelty-and-depth-register.md
- pyproject.toml
- src/bp_registry/artifacts.py

Phase 2 verification (continued)

  - src/bp_registry/canonical/__init__.py
  - src/bp_registry/canonical/budgets.py
  - src/bp_registry/canonical/contracts.py
  - src/bp_registry/canonical/execution.py
  - src/bp_registry/canonical/kernel.py
  - src/bp_registry/canonical/lineage.py
  - src/bp_registry/canonical/metrics.py
  - src/bp_registry/canonical/numerics.py
  - src/bp_registry/canonical/problems.py
  - src/bp_registry/canonical/profiling.py
  - src/bp_registry/canonical/schedules.py
  - src/bp_registry/canonical/tracing.py
  - src/bp_registry/canonical/verification.py
  - src/bp_registry/cli.py
  - src/bp_registry/css.py
  - src/bp_registry/evaluator.py
  - src/bp_registry/ledger.py
  - src/bp_registry/manifests.py
  - src/bp_registry/reference.py
  - src/bp_registry/runner.py
  - src/bp_registry/sampling.py
  - tests/canonical/conftest.py
  - tests/canonical/test_budgets.py
  - tests/canonical/test_contracts.py
  - tests/canonical/test_flooding.py
  - tests/canonical/test_numerics.py
  - tests/canonical/test_policy_evidence.py
  - tests/canonical/test_profiling.py
  - tests/canonical/test_tracing.py
  - tests/canonical/test_truth_boundary.py
  - tests/canonical/test_work_accounting.py
  - tests/fixtures/canonical-semantic-events.json
  - tests/test_canonical_cli.py
  - tests/test_ledger.py
  - tests/test_reference_smoke.py
covered_digest: "v1:sha256:aad66fe38159cfbec2e29e2a852fa8bda2ce55e91e6f061e038fa4152c9c298b"
behavior_unverified: 0
overrides_applied: 0
prohibitions_verified: 3
re_verification:
  previous_status: gaps_found
  previous_score: 2/5
  gaps_closed:
    - "CR-01: public preflight now admits only the implemented seven-role baseline and records active
    versus identity-only effects"
    - "CR-02: exact hierarchical work limits now reach reserve-before-mutation execution"
    - "CR-03: every admitted BP attempt now commits exactly one bp_calls unit"
    - "CR-04: every structural append now requires live shot ownership"
    - "WR-01 through WR-07: profiling cleanup, cohort identity, public profile/trace admission, syndrome
     documentation, candidate validation/current lineage, and native import isolation"
  gaps_remaining: []
  regressions: []

Phase 2: Canonical CSS Execution and Policy Contracts Verification Report

Phase 2 verification (continued)

Phase Goal: As a researcher, I want to inspect and run a canonical binary BP policy composition, so that I can trust its outcomes and computational attribution.
Verified: 2026-09-24T01:13:44Z
Status: passed
Re-verification: Yes — after Waves 8–14 closed the prior verification and review gaps

User Flow Coverage

User story: “As a researcher, I want to inspect and run a canonical binary BP policy composition, so that I can trust its outcomes and computational attribution.”

Step	Expected	Current codebase evidence	Status
Preflight the checked manifest	The researcher sees an immutable artifact, named numerical profile, seven policy bindings, scoped work limits, and transparent development labels before execution	`preflight_canonical()` in `src/bp_registry/runner.py`; strict `CanonicalExperimentV2` in `src/bp_registry/manifests.py`; current preflight contract test	✓ VERIFIED
Inspect the composition	Every policy role has a closed kind, binding identity, lifecycle/state contract, and active or identity-only public effect	`src/bp_registry/canonical/contracts.py`; baseline admission/effect functions; current checked `verify-run` output	✓ VERIFIED
Run the canonical experiment	The public command samples a CSS observation, runs project-owned flooding SPA, freezes selection, evaluates independently, and writes two terminal shots	`run_canonical_one()` → `run_canonical_structural_shot()` → `BinarySpaKernel`; current tutorial and public-command tests passed	✓ VERIFIED
Inspect attribution	Per-shot structural events expose attempt/candidate lineage, qualified iterations, atomic work, runtime/profile facts, and missingness	`query_canonical_events()`; checked ledger has seven events per shot, one `bp_calls` and one check-to-variable update per attempt, and enabled runtime evidence	✓ VERIFIED
Verify the outcome	The researcher can reject altered provenance/work/policy evidence and reconstruct the checked result from the ledger	`validate_canonical_shot()` and `verify_run()`; current checked command verified two rows; rehashed tampering tests passed	✓ VERIFIED
Outcome	Outcomes and computational attribution are independently checkable rather than trusted from the result JSON alone	Full public identity, deterministic attempt reconstruction, evaluator replay, exact event grammar, and ledger hash checks converge in `verification.py` and `cli.py`	✓ VERIFIED

The 2026-09-23 UAT remains historical corroboration only. The verdict above comes from current source inspection, current focused behavior tests, and fresh verification of the checked evidence after the remediation waves.

Goal Achievement

Observable Truths

#	Roadmap success criterion	Status	Evidence
1	A manifest runs project-owned sum-product BP with flooding from CSS observations through the independent evaluator, ledger, and small result artifact.	✓ VERIFIED	`runner.py` samples outside the engine and calls `run_canonical_structural_shot`; `execution.py` uses `BinarySpaKernel`/`FloodingSchedule`, freezes a candidate, then calls the independent evaluator. The public end-to-end/tutorial tests passed, and the checked two-row result passes `verify-run`.
2	Typed lifecycle/compatibility, immutable originals, truth isolation, numerical conventions, message order, and scoped stopping are enforced; invalid compositions and truth access are prevented.	✓ VERIFIED	Closed seven-role models and runtime-owned decisions are in `contracts.py`/`execution.py`; immutable problem/candidate types are in `problems.py`; stable finite SPA and double buffering are in `numerics.py`/`schedules.py`; exact hierarchical reservations are in `budgets.py`. Current lifecycle, hostile truth/candidate, rollback, baseline-admission, and scoped-denial tests passed.

Phase 2 verification (continued)

3	Deterministic small cases verify update equations and policy transitions independently of curves, with a reusable source-to-specification/fixture pattern.	✓ VERIFIED	Numerical box-plus, finite-boundary, flooding snapshot, lifecycle, lineage, policy-evidence, tracing, and work-reduction tests exercise small deterministic cases. `PolicyEvidence` requires a primary source/procedure and ordered invariants. No curve result is used as algebraic proof.
4	Control-flow/candidate events and work/runtime are queryable by shot and attempt; unobservable native counters retain missingness reasons.	✓ VERIFIED	`query_canonical_events()` filters validated v2 evidence by run/shot/kind; attempt IDs persist through attempt/candidate/selection/terminal data; `WorkVector` and profile evidence are typed. A fresh query returned seven events for each checked shot with distinct attempt IDs and `bp_calls=1`. Native availability/zero separation and structural writer ownership tests passed.
5	A researcher can install, run the tutorial, and inspect conventions, lifecycle contracts, and compatibility examples.	✓ VERIFIED	`docs/canonical-engine.md` contains the no-dependency editable-install instruction, public preflight/canonical/reconcile/verify flow, corrected `s_Z = H_Z e_X (mod 2)` convention, policy/lifecycle tables, compatibility rules, work/lineage semantics, and explicit evidence limits. `test_documented_tutorial` executes the flow from a temporary directory and passed.

Score: 5/5 truths verified (0 present-but-behavior-unverified)

All runtime-transition and ordering claims used for the score have current behavioral coverage. None was promoted from symbol presence alone.

Re-verification of Prior Gaps

Prior item	Current closure evidence	Status
CR-01 policy dispatch	Public preflight rejects nonbaseline executable kinds and materializes exact active/identity-only effects; the corresponding focused test passed.	Closed
CR-02 scoped budgets	Exact `WorkVector` limits flow from the public manifest into `BudgetTree`; reserve-before-sweep and scoped denial tests passed.	Closed
CR-03 BP-call accounting	`run_canonical_structural_shot()` reserves and commits `WorkVector(bp_calls=1)` once per admitted attempt; checked rows and current test report one call.	Closed
CR-04 shot ownership	`CanonicalEventWriter` requires a live owner for every non-begin append and reserves lifecycle kinds; the hostile second-writer test passed.	Closed
WR-01 profiling cleanup	Exceptional owned/caller-owned `tracemalloc` tests exist; current source retains cleanup in the profiler-owned `finally` path.	Closed
WR-02 replay cohort binding	`ReplayLink`/`CohortSelection` bind sample/run identity and reject mismatched cohorts.	Closed
WR-03 profile/trace admission	The supported clean low-overhead profile is honored; unbounded/nonclean and enabled public tracing reject before ledger creation.	Closed
WR-04 syndrome convention	The tutorial now states `s_Z = H_Z e_X (mod 2)` and its current contract test passed.	Closed
WR-05 candidate coercion	Canonical candidate payload validation rejects booleans, fractional/nonvector, and nonbinary values before content addressing.	Closed
WR-06 current lineage	Working-parent edges and current attempt/candidate identity are reconstructed and checked before evaluation and reuse.	Closed
WR-07 native isolation	Canonical preflight/run lazily isolates `ldpc`; canonical execution no longer imports the optional native adapter path.	Closed

Advisory (New Scope, Unevidenced)

None. The GSD decision-ID coverage heuristic emitted a nonblocking warning because shipped code/docs do not repeat 24 `D-xx` identifiers verbatim; the actual decision content was checked directly in contracts, runtime wiring, tests, and documentation. This is not a goal gap.

Phase 2 verification (continued)

Required Artifacts

The deterministic artifact query passed all **58/58** artifacts declared across Plans 02-01 through 02-16. Substantive/manual inspection additionally checked these functional groups.

Artifact group	Expected	Status	Details
Canonical schema and development manifest	Strict public schema and runnable seven-role fixture	✓ VERIFIED	Generated schema matches the model; preflight verifies artifact/environment/profile/budgets before sampling.
`contracts.py`, `execution.py`, `problems.py`	Policy/lifecycle boundary, immutable problem lineage, engine execution	✓ VERIFIED	Substantive implementations; public runner uses them and hostile tests exercise the boundary.
`numerics.py`, `kernel.py`, `schedules.py`	Project-owned finite binary SPA and complete flooding sweep	✓ VERIFIED	No native decoder delegation; deterministic equation/order tests passed.
`budgets.py`, `metrics.py`, `profiling.py`	Exact work, scoped limits, timing/resource facts, explicit missingness	✓ VERIFIED	Reservations and typed measurements feed runtime/work/terminal evidence; transition tests passed.
`lineage.py`, `tracing.py`, `verification.py`	Stable lineage, bounded optional traces, independent structural verification	✓ VERIFIED	Current candidate/attempt IDs and exact terminal grammars are enforced against rehashed tampering.
`manifests.py`, `runner.py`, `ledger.py`, `cli.py`	Public preflight/run/evidence/query/result wiring	✓ VERIFIED	These form the actual CLI path; checked and temporary runs pass.
Checked `events.jsonl` and `result.json`	Immutable checked evidence	✓ VERIFIED	`verify-run` returned two verified rows; SHA-256 values were recorded and `git diff --exit-code` showed no tracked-byte change.
Phase 2 docs and canonical tests	Executable tutorial, evidence boundaries, independent small cases	✓ VERIFIED	Documentation tests, CLI tutorial, and focused behavioral tests passed.

Key Link Verification

The mechanical key-link query found **38/39** links. Its sole reported miss was a parser limitation: Plan 02-16 used `src/bp_registry/canonical/profiling.py::LocalProfiler._peak_rss` as a source selector rather than a file path. Manual inspection verifies that link, making the substantive result **39/39**.

From	To	Via	Status	Details
Public manifest	`preflight_canonical()`	Strict parse, artifact/environment checks, baseline admission	✓ WIRED	Happens before sample, ledger, or kernel work.
`run_canonical_one()`	canonical engine	Resolved code/observation, compact policy manifest, work/profile controls	✓ WIRED	Public path calls `run_canonical_structural_shot()` directly.
Canonical engine	independent evaluator	Frozen current-attempt original-space candidate	✓ WIRED	Hidden truth appears only at the outer evaluator call after selection.
Engine work/profile	structural ledger	Owner-held attempt/runtime/candidate/selection/work/termination events	✓ WIRED	Same committed work summary feeds work and terminal records.
Structural ledger	result/`verify-run`	Exact grammar, public identity, independent reconstruction, ledger SHA-256	✓ WIRED	Rehashed provenance/work/policy tampering tests passed.
Native adapter	metric queries	Typed observed/unavailable `MetricFact` evidence	✓ WIRED	Unavailable counters remain `None` with reasons, distinct from observed zero.
`LocalProfiler._peak_rss()`	public profile validator	Exact producer-reachable method/reason/raw-context grammar	✓ WIRED (manual)	Source and focused unavailable-RSS producer test confirm the Plan 02-16 link.
Tutorial	manifest/CLI/result	Copyable commands and checked expected fields	✓ WIRED	`test_documented_tutorial` executes all four public commands.

Phase 2 verification (continued)

Data-Flow Trace (Level 4)

Value/evidence	Source	Flow	Produces real data	Status
CSS observation	Immutable artifact + deterministic `sample_shot()`	Preflight → decoder-visible syndrome → engine	Yes	✓ FLOWING
BP candidate	Channel prior + project-owned flooding updates	Numerics/schedule → frozen original-space candidate → selection	Yes	✓ FLOWING
Outcome	Hidden sampled error retained outside decoder	Frozen selected candidate + original code/truth → evaluator → terminal	Yes	✓ FLOWING
Work attribution	Leaf reservations and complete committed sweeps	`BudgetTree` → attempt summary → work and terminal events	Yes	✓ FLOWING
Runtime/profile	Live local profiler measurement	`LocalProfiler.measure()` → runtime/terminal copies → verifier grammar	Yes, with typed missingness	✓ FLOWING
Result counts	Validated terminal ledger rows	Terminal rows → outcome/termination counters → `result.json`	Yes	✓ FLOWING

No rendered or reported Phase 2 value terminates in a mock/static fallback. Development labels and the two-shot budget are intentionally fixed manifest facts, not population measurements.

Behavioral Spot-Checks

Behavior	Command	Result	Status
Public tutorial, structural evidence, admission, BP-call, scoped denial	Five named `test_canonical_cli.py` tests under `timeout 10s`	5 passed in 5.42s	✓ PASS
Lifecycle ownership, truth/current-candidate boundary, rollback invariants	Five named contract/truth/budget tests	6 parameterized cases passed in 0.37s	✓ PASS
SPA equation, flooding ordering, candidate retention, work reduction, RSS grammar, evidence limits	Six named canonical tests	6 passed in 0.35s	✓ PASS
Native missingness/zero distinction, structural ownership, sparse evidence	Four named ledger/reference/tracing tests	4 passed in 3.37s	✓ PASS
Rehashed work/policy and provenance tampering	Two named CLI tests	3 parameterized cases passed in 4.59s	✓ PASS
Honest checked labels and no-promotion documentation	Two named tests	2 passed in 0.92s	✓ PASS
Checked evidence reconstruction	`python -m bp_registry.cli verify-run data/results/canonical-three-qubit-development`	Two terminal rows verified	✓ PASS
Query by shot/attempt	`query_canonical_events()` over checked ledger	Seven events per shot; distinct attempt IDs; `bp_calls=1`; runtime enabled	✓ PASS

The current security report independently records the full workspace run as **180 passed**. This verification did not use that headline as sole proof; it ran the focused behaviors above and inspected their implementation paths.

Probe Execution

No phase-declared or conventional `scripts/**/tests/probe-*.sh` probe exists. **Step 7c: N/A**; executable verification is provided by named tests and the public CLI command.

Requirements Coverage

Requirement	Source plans	Status	Current evidence
ENG-01	02-01, 02-08, 02-10–16	✓ SATISFIED	Public runner uses `BinarySpaKernel`, not native decode; tutorial/result verification passed.
ENG-02	02-02, 02-10	✓ SATISFIED	Closed registry/composition/runtime decision checks plus

Phase 2 verification (continued)

baseline public admission test. |
ENG-03	02-04, 02-10, 02-13–15	✓ SATISFIED	Exact work resources, ancestor reservation, terminal budget evidence, public denial test.
ENG-04	02-03, 02-09, 02-12	✓ SATISFIED	Named profile, stable SPA tests, double-buffered flooding, corrected tutorial.
ENG-05	02-01, 02-02, 02-05, 02-11, 02-13–15	✓ SATISFIED	Hostile capability/candidate tests; frozen selection precedes evaluator; replay rederives lineage.
ENG-06	02-02, 02-03, 02-09	✓ SATISFIED	`PolicyEvidence` enforcement, docs anchors, small-case contract/numerical tests.
SCH-01	02-03, 02-04	✓ SATISFIED	Canonical edge table, old snapshot/fresh buffer, exact complete-sweep work tests.
LED-02	02-05, 02-08, 02-11–15	✓ SATISFIED	Owned structural event grammar, lineage, bounded trace implementation, public trace-off admission.
LED-03	02-04, 02-07, 02-08, 02-10–16	✓ SATISFIED	`WorkVector`, qualified iterations, `bp_calls=1`, runtime profile, no universal scalar.
LED-04	02-06, 02-08	✓ SATISFIED	Typed metric facts and availability queries; zero-vs-unavailable tests passed.
DOC-01	02-09, 02-12	✓ SATISFIED	Executable tutorial passed; docs cover equations, lifecycle, compatibility, attribution, evidence limits.

No Phase 2 requirement is orphaned: the roadmap and `REQUIREMENTS.md` map exactly these eleven IDs, and every ID appears in Phase 2 plan frontmatter.

Prohibition and Manual-Assumption Resolution

The three legacy prohibition records omit the newer `verification: test|judgment` key and retain `status: unverified, flagged: true`. They are not silently treated as green. Each statement is objectively enforced on the current Phase 2 claim surfaces, so it is resolved with wired test evidence; historical human UAT is corroboration, not the sole basis.

Item	Current enforcement	Historical human evidence	Verdict
PROH-LED-03-01 — no outcome-conditioned/rare-event population estimate without prespecified probabilities/weighting	Profiling strategy states the negative rule; policy-evidence test requires it; typed cohorts retain pre-outcome selection and inclusion probability	UAT Test 6 explicitly passed the representative-vs-diagnostic/population boundary	✓ VERIFIED
PROH-ENG-01-01 — no source-faithful/statistically assessed label for the development fixture	Manifest, terminal/result, CLI, and verifier require `canonical_implemented`, `development_fixture`, `not_assessed`; honest-label test and fresh `verify-run` passed	UAT Tests 1, 14, and 15 exercised the labeled evidence flow	✓ VERIFIED
PROH-DOC-01-01 — no statistical-reproduction or established-novelty claim for development/profile/trace evidence	All three Phase 2 docs state the negative boundary; policy-evidence tests require “not a reproduction claim” and “not established methodological novelty”	UAT Test 6 explicitly passed the no-promotion wording	✓ VERIFIED

Manual assumptions are resolved for the current Phase 2 surface: UAT Test 4 covered scope/reason interpretability, Test 5 covered mathematical/lifecycle readability, and Tests 10–11 covered native missingness. Later remediation was regression-checked by current cap, tutorial, policy-evidence, and native-missingness tests. The LED-04 note about a hypothetical future backend is not a present Phase 2 behavior and remains a future verification obligation if such a backend is added.

Anti-Patterns Found

File	Line/pattern	Severity	Impact
`docs/canonical-engine.md`	`XXXXXX` in the documented `mktemp` template	ℹ Info / false positive

Phase 2 verification (continued)

 Secure temporary-directory template syntax, not a debt marker. |

No `TBD`, `FIXME`, or `XXX` debt marker, skipped/xfail test, placeholder implementation, console-only handler, or user-visible hardcoded-empty stub was found. Empty-list/dict returns are legitimate empty-file or incomplete-optional-context guards, not fake successful evidence.

Human Verification Required

None. The CLI user flow is exercised end to end in a disposable directory, and the prior human judgments have directly matching current regression evidence. There is no visual UI, external service, or unexercised state-transition invariant left for this phase.

Gaps Summary

No remaining gaps. All five roadmap truths and all eleven Phase 2 requirements are verified against current code and behavior. The checked evidence remains honestly scoped as a two-shot canonical development fixture, not a paper reproduction or statistical assessment.

Verified: 2026-09-24T01:13:44Z
Verifier: the agent (gsd-verifier)